

Matthew Jelinek

Aspiring network engineer and current student at Southwest Wisconsin Technical College.
Reliable and loyal worker who wants to see the world be better.

> [Linkedin Profile](#)

> [Github Portfolio](#)

Experience

Core Systems

- Windows 10/11
- Cisco Appliances
- Linux
- Windows Server 2016

Hardware

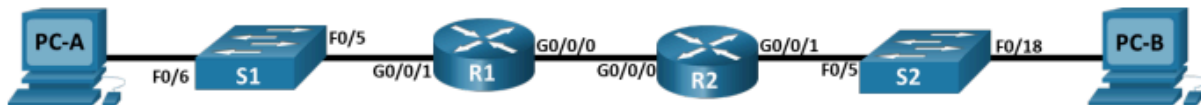
- RAM replacement
- Storage Upgrades
- Troubleshooting Hardware
- Building Networks

Software/Clouds

- PuTTY
- Packet Tracer
- SQL
- HTML/JS/CSS

Scenario

In a simulated environment, we were asked to build a simple network with two routers, two switches and two host PCs. The topology is displayed in more detail below.



Addressing Table

Device	Interface	IP Address	Subnet Mask	Default Gateway
R1	G0/0/0	10.0.0.1	255.255.255.252	N/A
	G0/0/1	N/A	N/A	
	G0/0/1.100			
	G0/0/1.200			
	G0/0/1.1000	N/A	N/A	
R2	G0/0/0	10.0.0.2	255.255.255.252	N/A
	G0/0/1			
S1	VLAN 200			
S2	VLAN 1			
PC-A	NIC	DHCP	DHCP	DHCP
PC-B	NIC	DHCP	DHCP	DHCP

We were paired into groups of two and instructed to fill out the addressing table displayed above and build and configure the network to support inter-VLAN routing and DHCPv4. I was

instructed to work on the network containing PC-A and my partner was to work on the side with PC-B.

First, calculating a subnet was required. Subnet A needed to support 58 hosts, Subnet B needed 28, and C only needed 12. With the initial network of 192.168.1.0/24, this means our subnets are as follows: 192.168.1.0/26, 192.168.1.64/27, and 192.168.1.96/28.

Wiring my side was simple, as was basic configuration: I keep a .txt file on a thumb drive to easily paste in basic configuration for routers and switches. Connecting the two sides required a longer ethernet cable than what we normally work with, but we managed to find one that only just spanned the distance between the two stacks.

I required three VLANs on my switch, 100, 200, and 1000, with 1000 being configured as the native VLAN. Trunking was configured between f0/5 on S1 and g0/0/1 on R1 and a default route was configured between R1 and R2 to allow for communication between the networks.

Education

- Currently Enrolled at Southwest Wisconsin Technical College as IT-Cybersecurity Specialist. Expected graduation in May 2027.
- Associates degree in Graphic and Web Design from Southwest Wisconsin Technical College. Graduated May 2018.
- High School Diploma from Prairie du Chien High School. Graduated June 2015.